THE STRICT TRANSFORM ALONG THE FITTING IDEAL LOCUS IS LOCALLY GENERATED BY FEW SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The proof is a two-step translation and says so; both named steps are recorded here so a reader can see how much of the argument lives outside the counted span.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Fitting criterion for local generation by k sections. Lemma lemma-fitting-ideal-generate-locally, used at proof lines 9314-9316. Turns the claim into Fit_k(F') = O_{S'}.
- Blowup along a Fitting ideal in algebra. More on Algebra, Lemma more-algebra-lemma-blowup-fitting-ideal, used at proof lines 9317-9319. Is the algebra statement the translation lands on.
- Fitting ideals of a module. Section section-fitting-ideals, used at proof lines 9304-9305. Defines Z_k.

Source reference. Original lemma 0C3F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15190–15190. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0CZM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16282–16282. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 0C3C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15187–15187. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_S$-module. Let $Z_k \subset S$ be the closed subscheme cut out by $Fit_k(\mathcal{F})$, see Section 0C3C. Let $S' \to S$ be the blowup of S in Z_k and let $\mathcal{F}'$ be the strict transform of $\mathcal{F}$. Then $\mathcal{F}'$ can locally be generated by $\leq k$ sections.*

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Recall that $\mathcal{F}'$ can locally be generated by $\leq k$ sections if and only if $\mathrm{Fit}_k(\mathcal{F}') = \mathcal{O}_{S'}$, see Lemma 0C3F. Hence this lemma is a translation of More on Algebra, Lemma 0CZM. $\square$